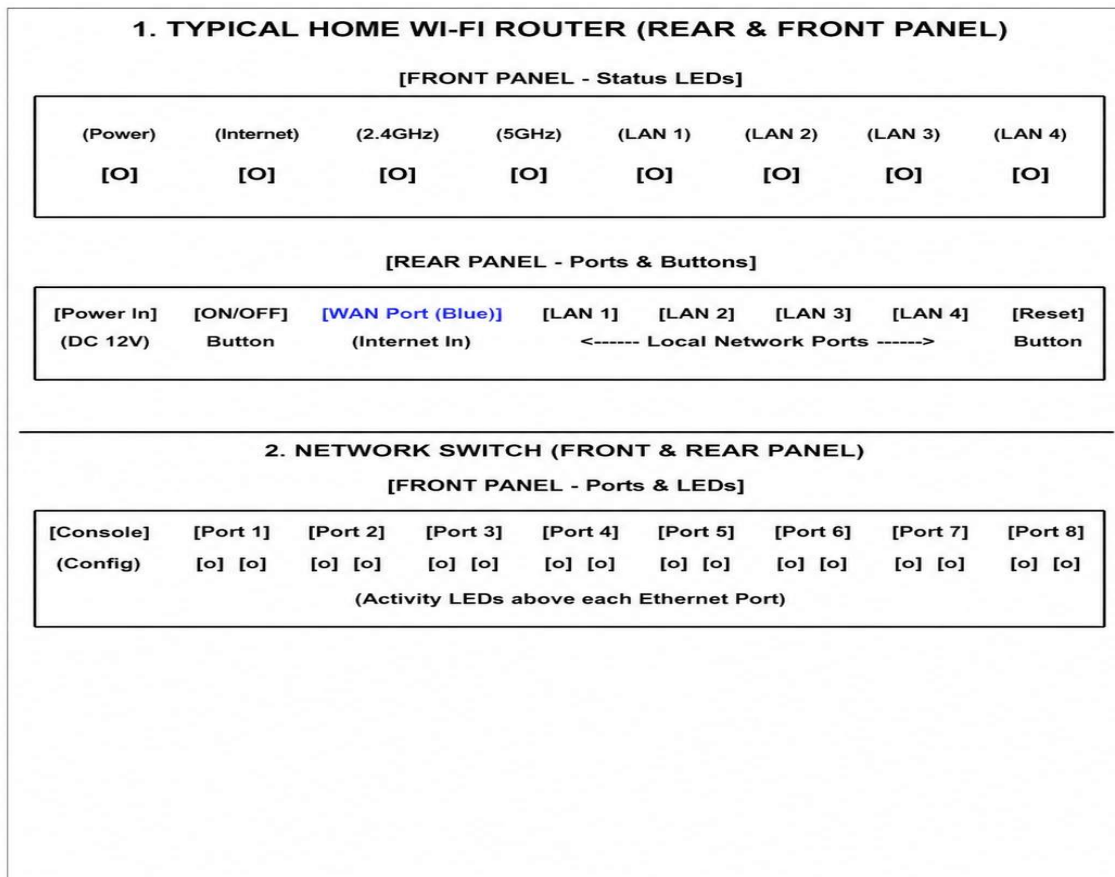


Task 1:

Home Wi-Fi Router and Network Switch Labelled Diagram Description



Objective: Illustrate and label the key physical ports and status LEDs on a typical home Wi-Fi router and an unmanaged/managed network switch.

1. Home Wi-Fi Router Components & Ports:

- **WAN Port (Wide Area Network / Internet Port):**

- *Color/Location:* Usually colored **Blue or Yellow** on the back panel and separated from LAN ports.
- *Purpose:* Connects directly to the ISP Modem via an Ethernet cable to bring external internet into the house.

- **LAN Ports (Local Area Network Ports):**

- *Color/Location:* Group of 4 RJ45 Ethernet ports on the back panel (usually Yellow/Black).
- *Purpose:* Connects local wired devices such as Desktop PCs, Laptops, Gaming Consoles, and Smart TVs.

- **Power Port (DC Input):**

- *Location:* Back panel next to the power switch.

- *Purpose:* Receives power from the 12V DC power adapter socket.

- **Status LEDs (Front Panel Indicators):**

- **Power LED:** Solid light indicates the router is powered ON.
 - **Internet/WAN LED:** Indicates active connection to the ISP. Blinking means data is transmitting.
 - **2.4GHz / 5GHz Wi-Fi LEDs:** Indicates wireless bands are active and broadcasting.
 - **LAN Status LEDs (1-4):** Lights up when a physical cable is plugged into the respective LAN port.
-

2. Network Switch Components & Ports:

- **Ethernet LAN Ports (Ports 1 to 8/16/24):**

- *Location:* Arranged in rows across the front panel.
- *Purpose:* Provides wired network connections to multiple devices, expanding the number of available physical network ports.

- **Console Port (On Managed Switches):**

- *Location:* Front or rear panel (RJ45 or Serial RS-232 interface).
- *Purpose:* Allows network administrators to connect a terminal PC to configure switch settings via CLI (Command Line Interface).

- **Power In Port:**

- *Location:* Rear panel of the switch casing.
- *Purpose:* Plugs into the main power cable or external power brick.

- **Link / Activity LEDs (Status Lights):**

- *Location:* Positioned directly above or integrated into each Ethernet port connector.
- *Purpose:* **Solid Green** indicates a successful physical link (100Mbps/1Gbps); **Blinking Green/Amber** indicates active data traffic passing through that port.

Task 2 :

Key Differences Between Routers and Switches

Objective: Compare and contrast the main functions, addressing mechanisms, and traffic handling techniques of Routers and Switches.

1. Operational OSI Layer & Addressing Type

- **Router: Operates at Layer 3 (Network Layer) of the OSI model. It relies on IP Addresses (Logical Addresses) to direct data packets across different subnets and networks.**
- **Switch: Operates at Layer 2 (Data Link Layer) of the OSI model. It relies on MAC Addresses (Physical**

Addresses) stored in its MAC address table to forward frames within the local network.

2. Network Scope & Connectivity

- Router: Designed to connect different/multiple networks together (e.g., connecting a Local Area Network [LAN] to the Wide Area Network [WAN/Internet]).**
 - Switch: Designed to connect multiple devices within the same single network (e.g., connecting PCs, printers, and servers inside one LAN).**
-

3. Data Handling & Traffic Routing Mechanism

- **Router:** Reads the destination IP address of incoming data packets and determines the optimal path (routing) to deliver them across external networks using routing tables and protocols (NAT/DHCP).
- **Switch:** Reads the destination MAC address of incoming data frames and directly forwards (switches) them only to the specific physical port connected to the recipient device, preventing network congestion.

Task 3 :

Router and Switch Component Identification & Description

Objective: Identify, describe positions, and state the purposes of the Power port, Ethernet ports, and LED status indicators on both a Wi-Fi Router and a Network Switch.



1. Wi-Fi Router Identification

- **Power Port (DC Input):**

- **Position:** Located on the far-left or far-right side of the rear panel.
- **Purpose:** Receives 12V DC power from the external wall power supply adapter to power on the device.

- **Ethernet Ports (WAN & LAN Ports):**

- **Position:** Located along the rear panel. Features 1 Blue/Yellow WAN port and 4 Yellow/Black LAN RJ45 ports.
- **Purpose:** The WAN port connects to the ISP modem for internet input; LAN ports connect local wired devices (PCs, Smart TVs, Laptops) using RJ45 network cables.

- **LED Status Indicators:**

- **Position:** Located on the top/front display panel.
- **Purpose:**
 - **Power LED:** Solid Green indicates power is ON.

- **Internet/WAN LED:** Solid/Blinking Green indicates active internet connectivity.
- **Wi-Fi LEDs (2.4GHz / 5GHz):** Blinking indicates active wireless data transmission.
- **LAN LEDs (1-4):** Solid Green indicates a physical cable is plugged in; blinking indicates active local data traffic.



2. Network Switch Identification

- **Power Port:**
 - **Position:** Located on the rear panel of the metal/plastic casing.

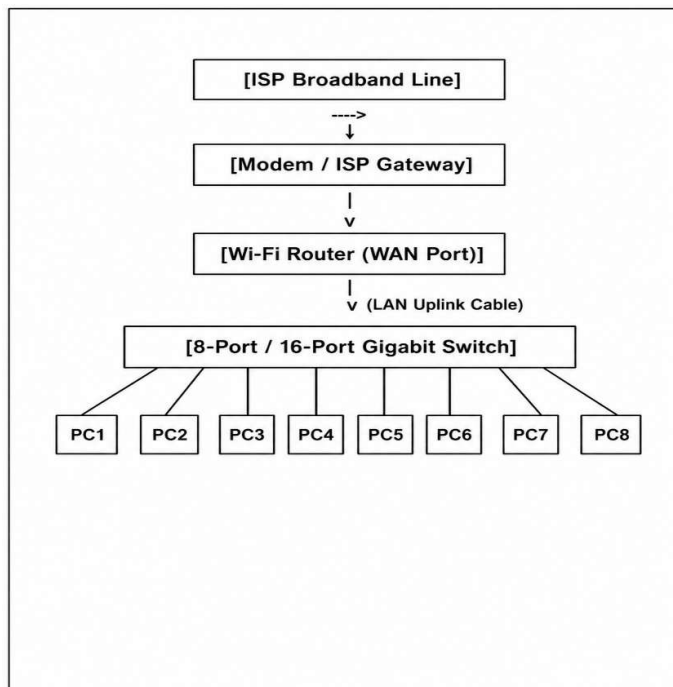
- **Purpose:** Plugs into the AC power cord or external DC power supply brick to power the internal switching hardware.
- **Ethernet Ports (Ports 1 to 8/16):**
 - **Position:** Arranged in numbered rows across the main front panel.
 - **Purpose:** Connects multiple network devices together inside a Local Area Network (LAN) to allow inter-device communication.
- **LED Status Indicators (Link/Activity LEDs):**
 - **Position:** Positioned directly above or built into the top corners of each RJ45 Ethernet port on the front panel.
 - **Purpose:**
 - **Solid Green:** Indicates an active physical link established with a connected device (e.g., 100Mbps / 1Gbps connection).
 - **Blinking Green/Amber:** Indicates real-time data packets are currently being transmitted/received through that specific port.

Task 4:

Gaming Café Network Setup Scenario & Justification

Objective: Design a reliable, high-speed network setup for an 8-PC gaming café handling multiplayer gaming and IPL live streaming.

1. Network Topology & Connection Flow:



2. Device Order & Configuration Steps:

1. First Device (Router to ISP Line):

- The **Router** will be connected directly to the ISP internet/modem line via its **WAN (Internet) port**.
- **Justification:** The router is required first because it handles Network Address Translation (NAT), acts as a hardware firewall, and runs a DHCP server to assign local IP addresses to all connected gaming devices.

2. Second Device (Switch to Router):

- An Ethernet cable will be connected from one of the router's **LAN ports** to Port 1 (Uplink Port) of an **8-Port/16-Port Gigabit Network Switch**.
- **Justification:** Routers typically only have 4 LAN ports, which is insufficient for 8 PCs. The switch expands the physical port capacity to connect all 8 gaming rigs seamlessly.

3. Connecting 8 Gaming PCs to the Network:

- Each of the 8 gaming PCs will be connected directly to the ports on the **Gigabit Switch** using **Cat6 Ethernet cables**.
- **Justification:** A wired switch connection provides dedicated 1 Gbps full-duplex bandwidth to every single PC with minimal ping jitter, low latency, and zero packet loss—essential for smooth multiplayer gaming and high-definition IPL live streaming.